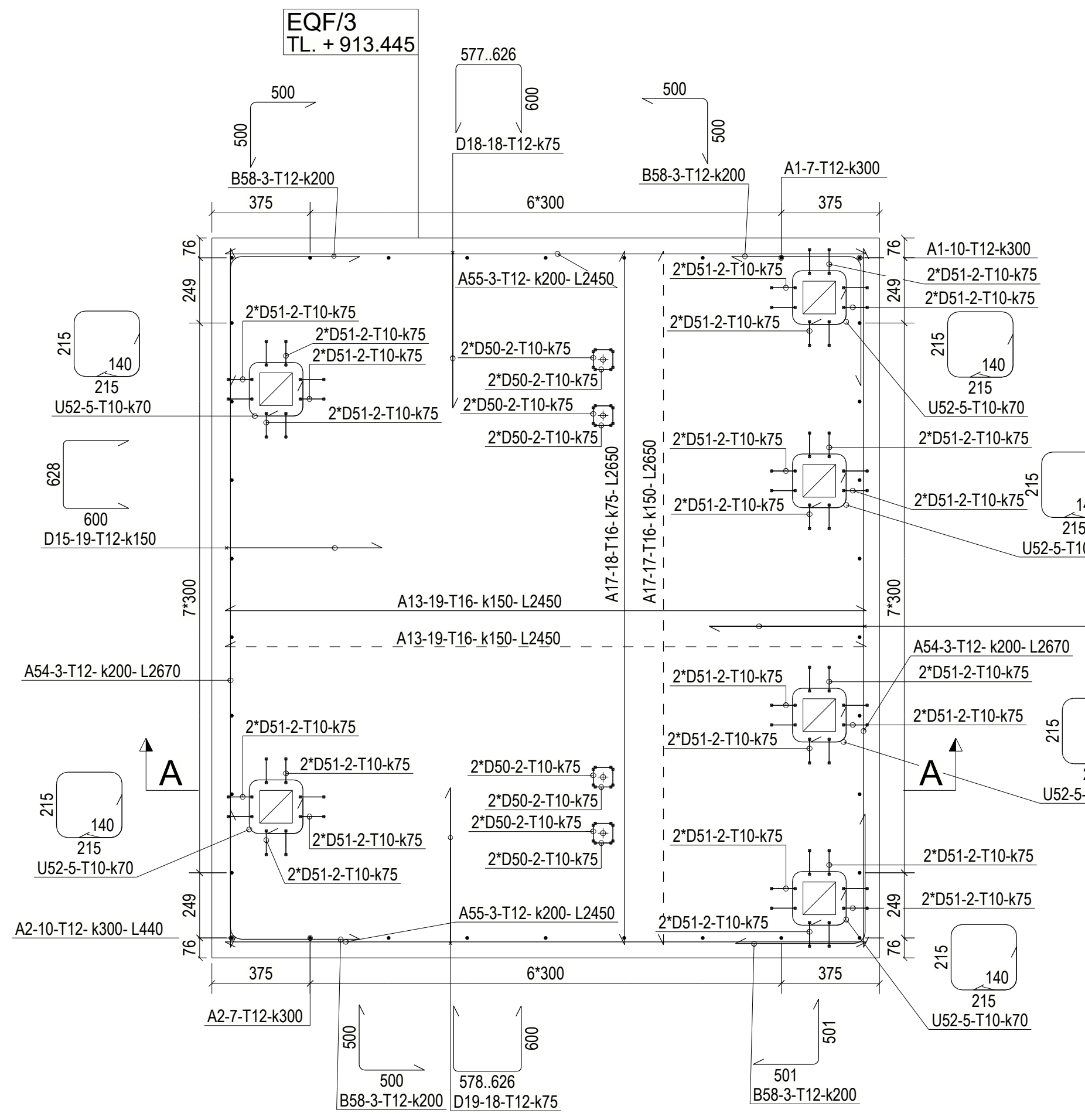
MD-216-3220-EG-CV-DED-1161 DRAWING NUMBER
REV D01

PUMPS FOUNDATION REINFORCEMENT PLAN

1:20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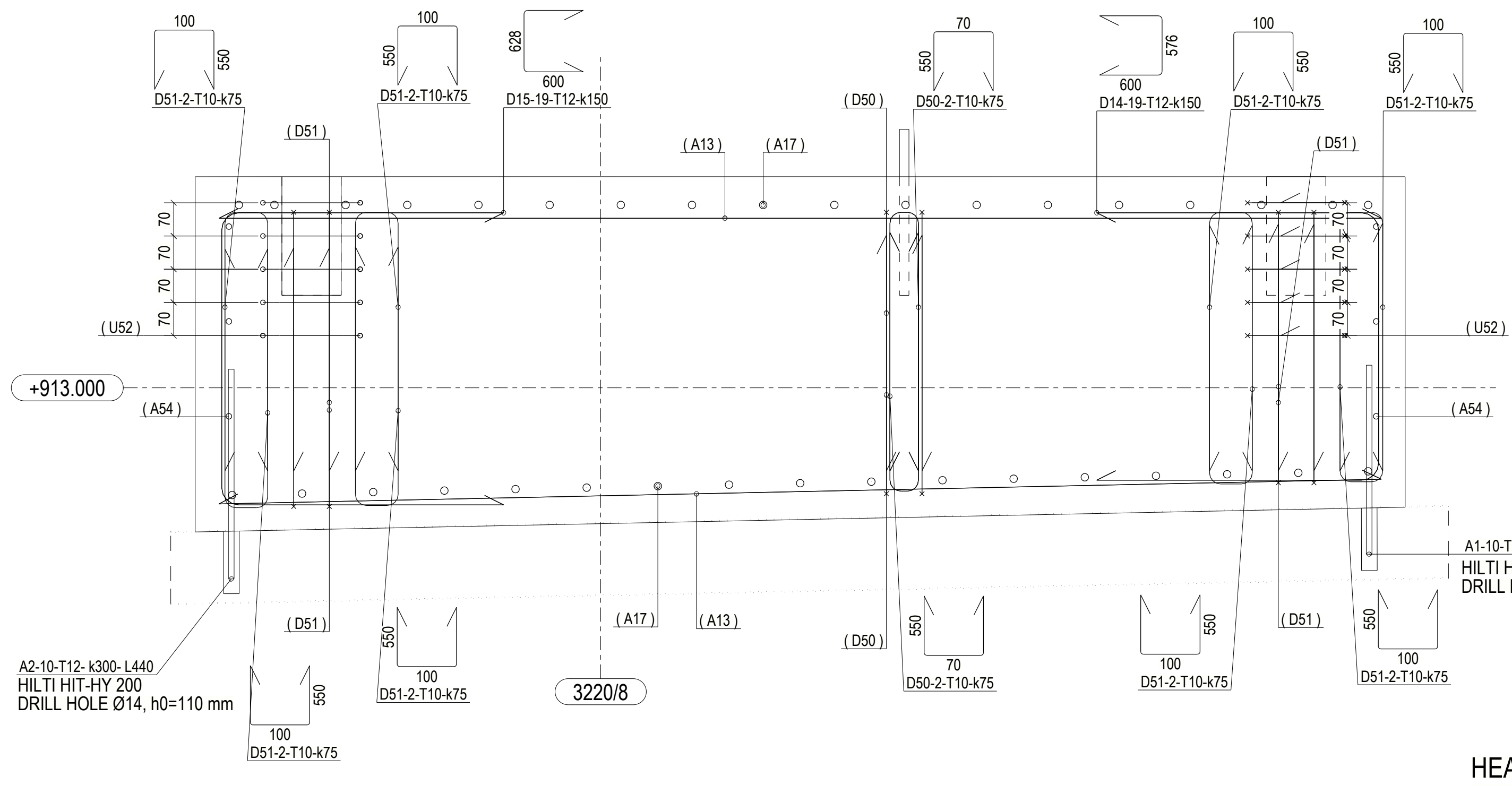
TL. + 913.445
B58-3-T12-k200
D18-18-T12-k75
A1-7-T12-k300
A1-10-T12-k300
2*D51-2-T10-k75
2*D50-2-T10-k75
A55-3-T12- k200- L2450
A13-19-T16- k150- L2450
A54-3-T12- k200- L2670
U52-5-T10-k70
D15-19-T12-k150
D14-19-T12-k150
A17-18-T16-k75- L2650
A2-10-T12- k300- L440
A2-7-T12-k300
B58-3-T12-k200
D19-18-T12-k75
6*300
7*300
A
A

A - A
1:10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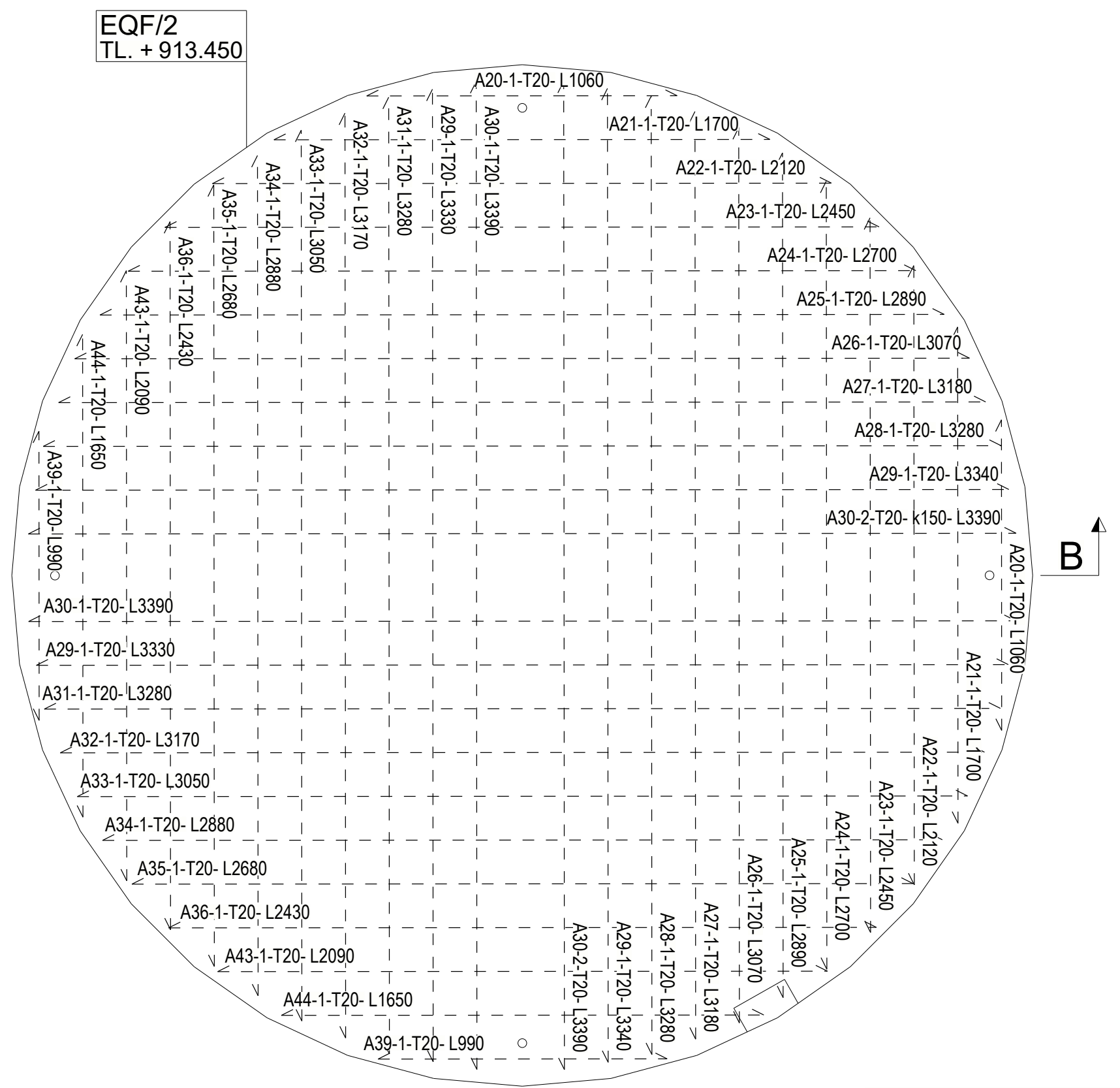
D15-19-T12-k150
D50-2-T10-k75
D14-19-T12-k150
(A13)
(A17)
(D50)
(D51)
(U52)
(A54)
+913.000
A1-10-T12- k300- L400
HILTI HIT-HY 200
DRILL HOLE Ø14, h0=110 mm
A2-10-T12- k300- L440
HILTI HIT-HY 200
DRILL HOLE Ø14, h0=110 mm
3220/8

WATER TANK FOUNDATION BOTTOM REINFORCEMENT

1:20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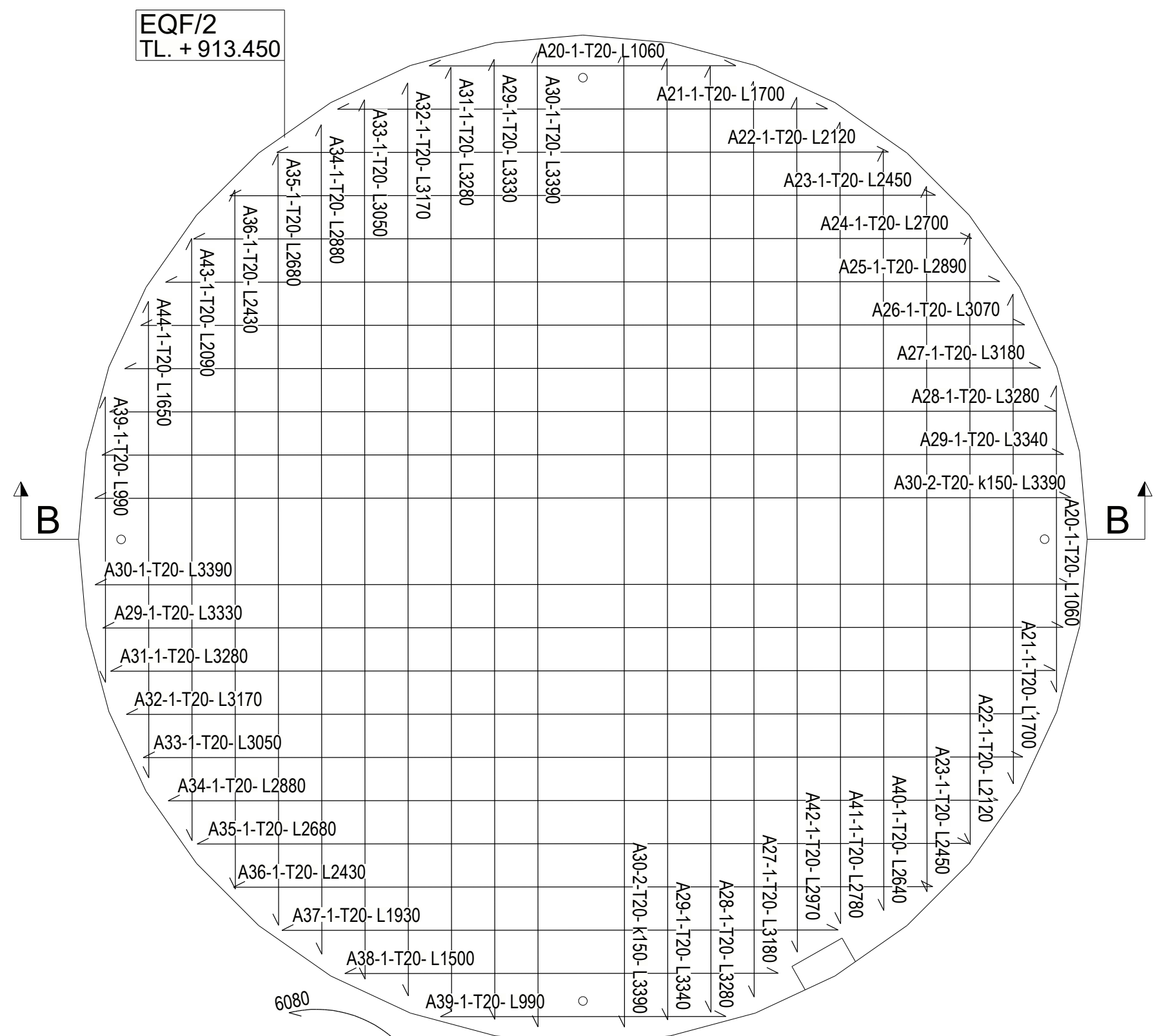
A20-1-T20- L1060
A21-1-T20- L1700
A22-1-T20- L2120
A23-1-T20- L2450
A24-1-T20- L2700
A25-1-T20- L2890
A26-1-T20- L3070
A27-1-T20- L3180
A28-1-T20- L3280
A29-1-T20- L3340
A30-2-T20- k150- L3390
A31-1-T20- L3280
A32-1-T20- L3170
A33-1-T20- L3050
A34-1-T20- L2880
A35-1-T20- L2680
A36-1-T20- L2430
A43-1-T20- L2090
A44-1-T20- L1650
A39-1-T20- L990
B
B

WATER TANK FOUNDATION TOP REINFORCEMENT

1:20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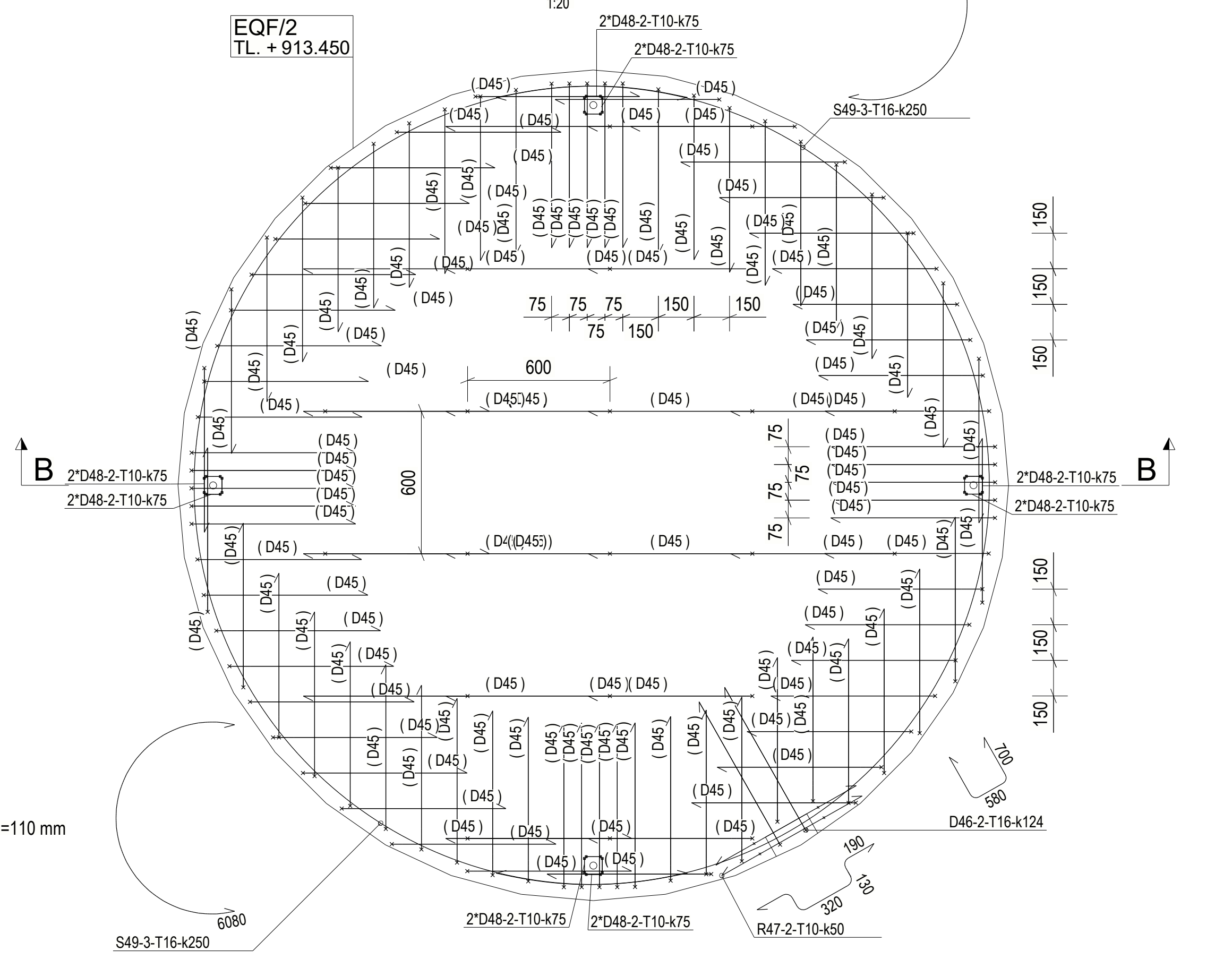
TL. + 913.450
A20-1-T20- L1060
A21-1-T20- L1700
A22-1-T20- L2120
A23-1-T20- L2450
A24-1-T20- L2700
A25-1-T20- L2890
A26-1-T20- L3070
A27-1-T20- L3180
A28-1-T20- L3280
A29-1-T20- L3340
A30-2-T20- k150- L3390
A30-1-T20- L3390
A29-1-T20- L3330
A31-1-T20- L3280
A32-1-T20- L3170
A33-1-T20- L3050
A34-1-T20- L2880
A35-1-T20- L2680
A36-1-T20- L2430
A37-1-T20- L1930
A38-1-T20- L1500
A39-1-T20- L990
6080
B
B

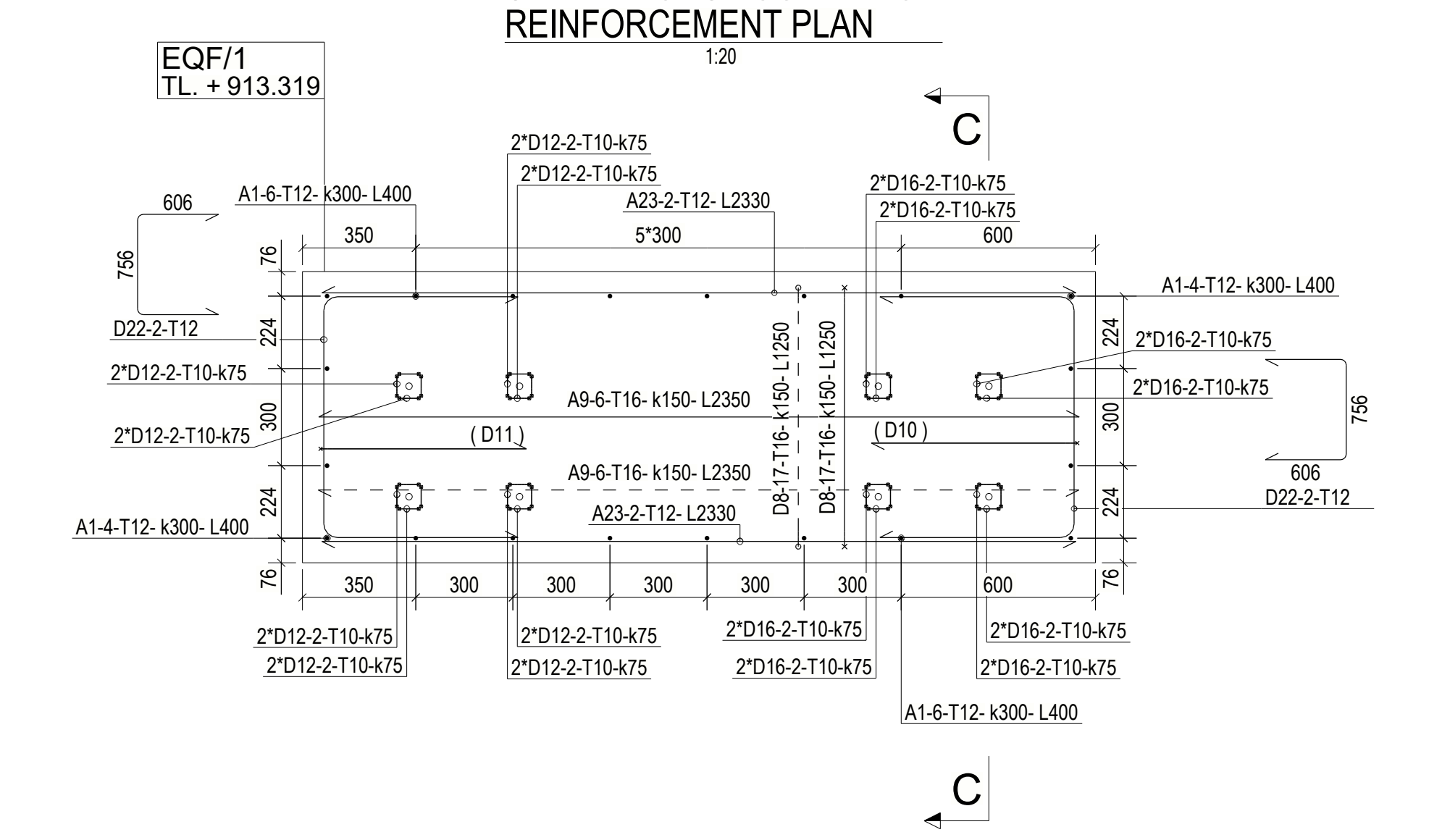
WATER TANK FOUNDATION EXTRA REINFORCEMENT

1:20


EQF/2
TL. + 913.450
2*D48-2-T10-k75
S49-3-T16-k250
S49-3-T16-k250
(D45)
600
75
150
D46-2-T16-k124
R47-2-T10-k50
6080
B
B

SEPARATORS FOUNDATION REINFORCEMENT PLAN

1:20


EQF/1
TL. + 913.319
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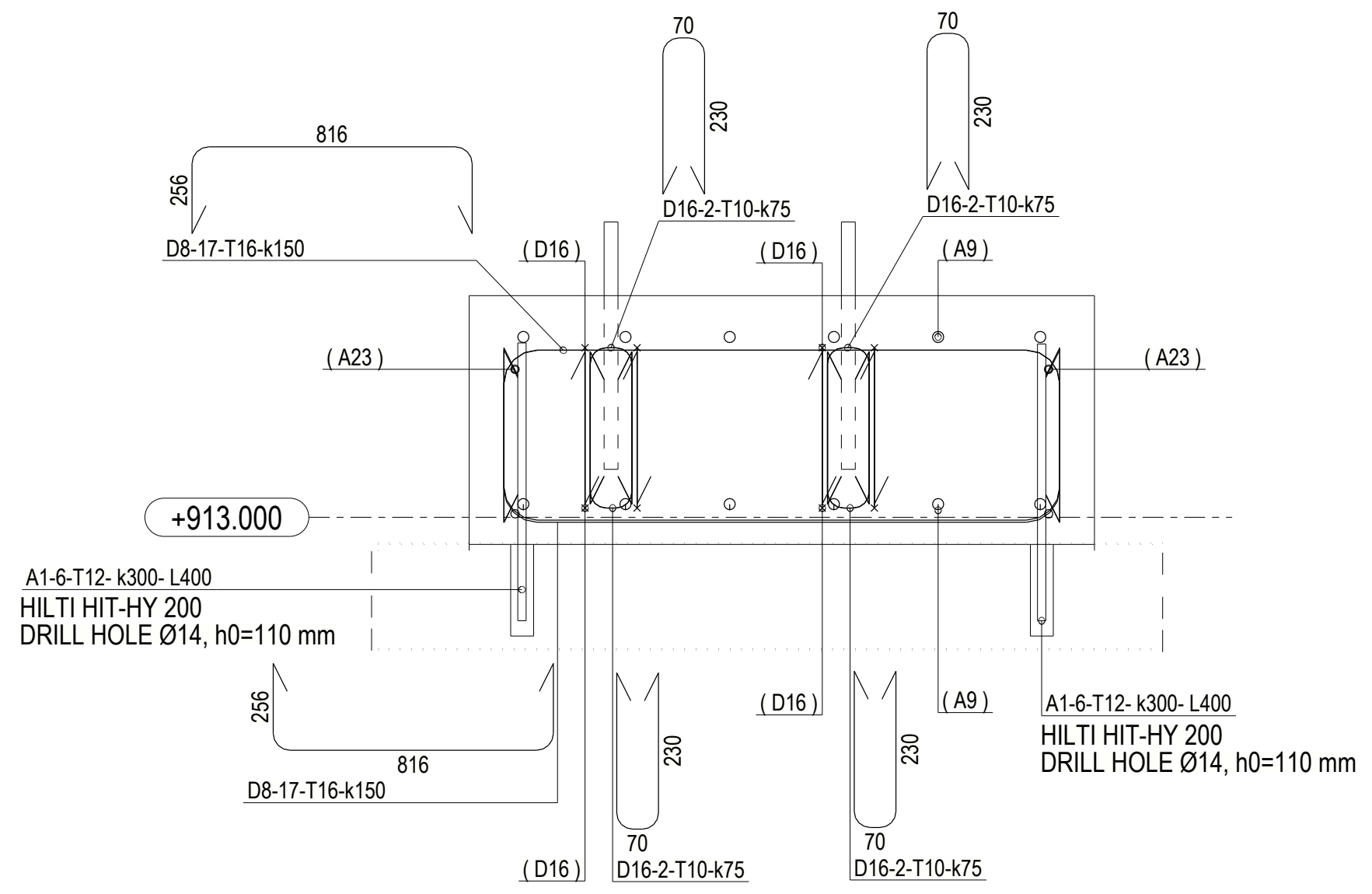
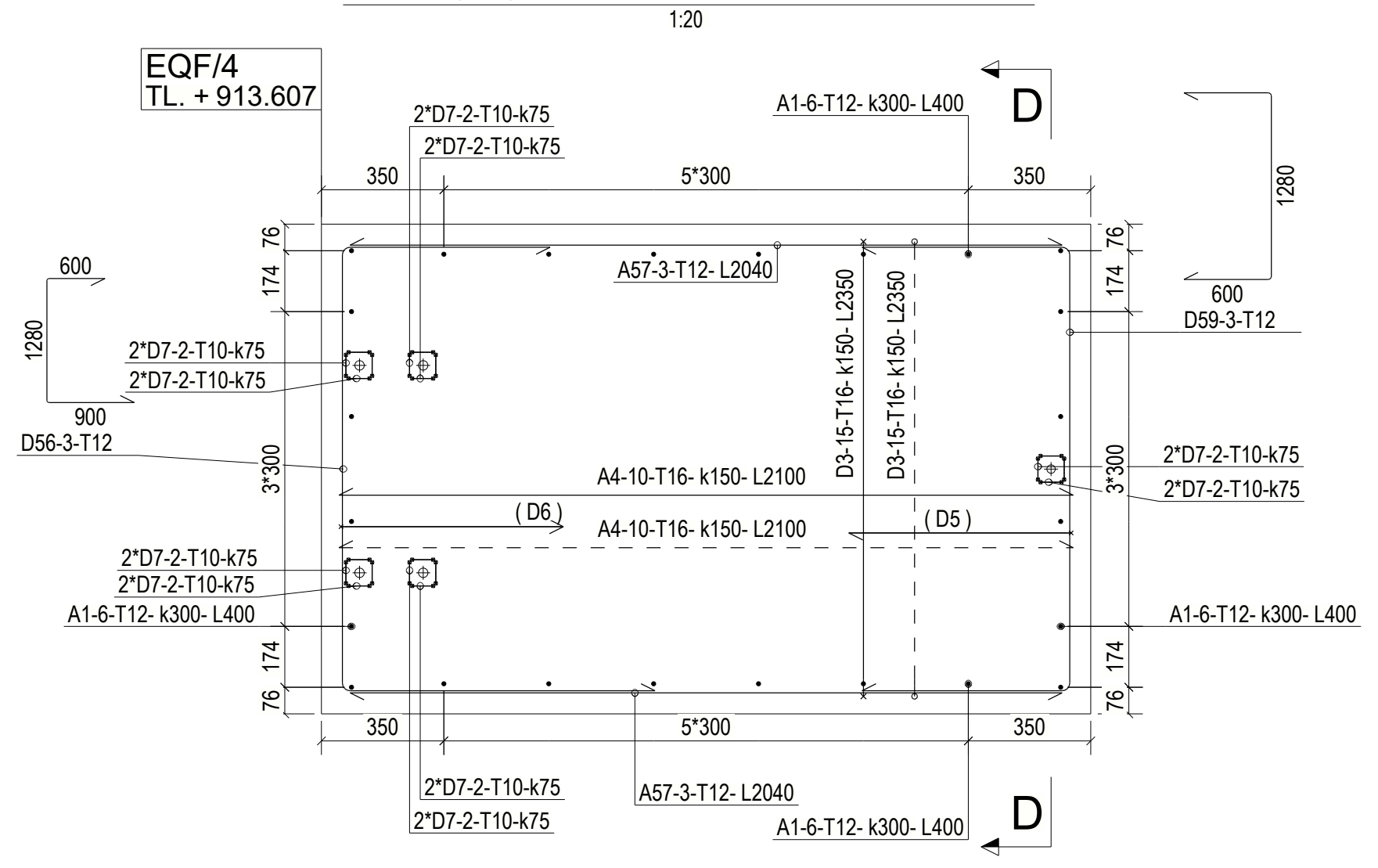
2*D12-2-T10-k75
A23-2-T12- L2330
2*D16-2-T10-k75
D22-2-T12
A1-4-T12- k300- L400
A9-6-T16- k150- L2350
(D11)
(D10)
D11-6-T12- L1250
D10-6-T12- L1250
A23-2-T12- L2330
A1-4-T12- k300- L400
A1-6-T12- k300- L400
5*300
C
C

C - C
1:10

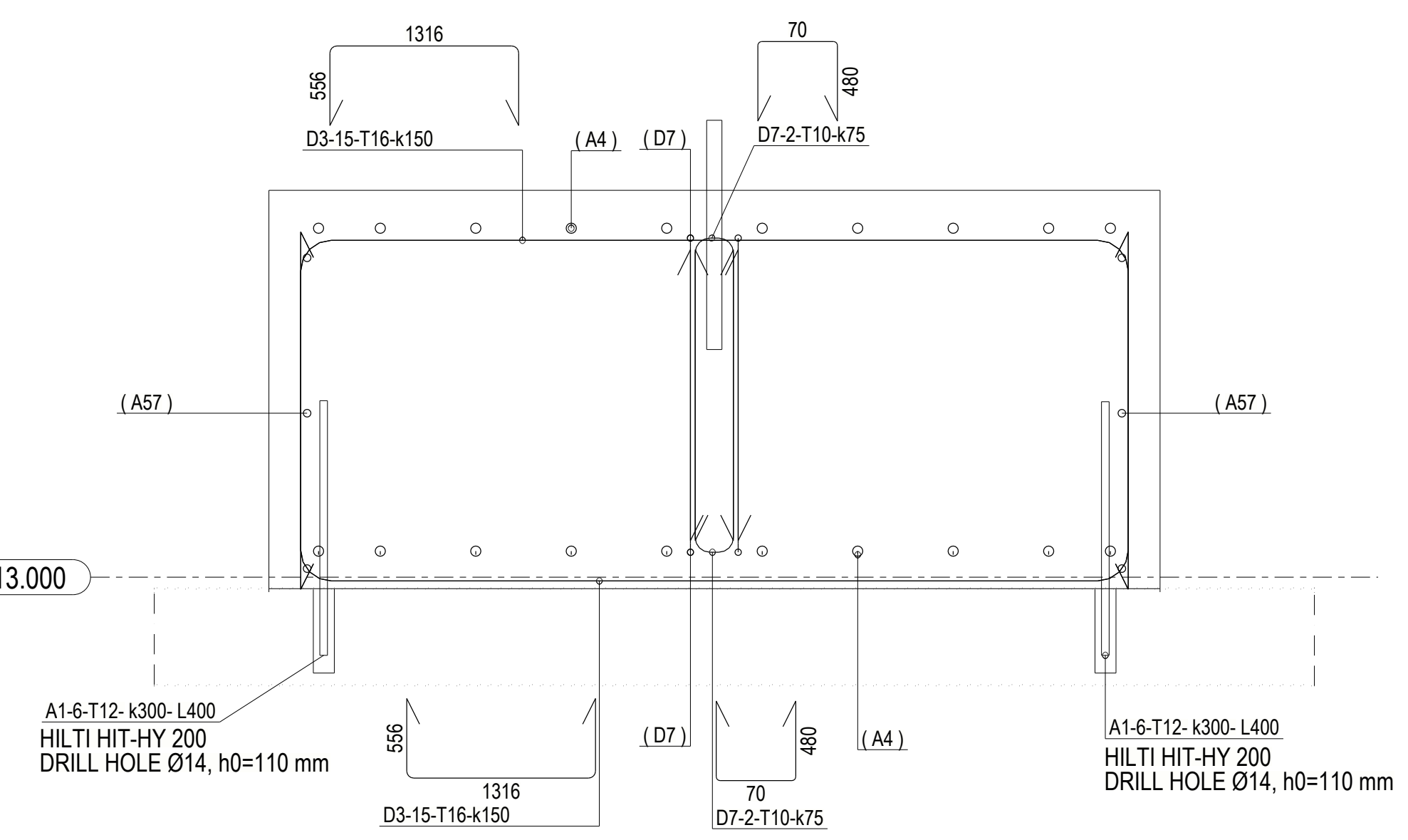

D8-17-T16-k150
D16-2-T10-k75
(D16)
(A9)
(A23)
+913.000
A1-6-T12- k300- L400
HILTI HIT-HY 200
DRILL HOLE Ø14, h0=110 mm
A1-6-T12- k300- L400
HILTI HIT-HY 200
DRILL HOLE Ø14, h0=110 mm
816
230
70
D8-17-T16-k150
D16-2-T10-k75
D16-2-T10-k75

HEAT EXCHANGER FOUNDATIONS, 2 pcs. REINFORCEMENT PLAN

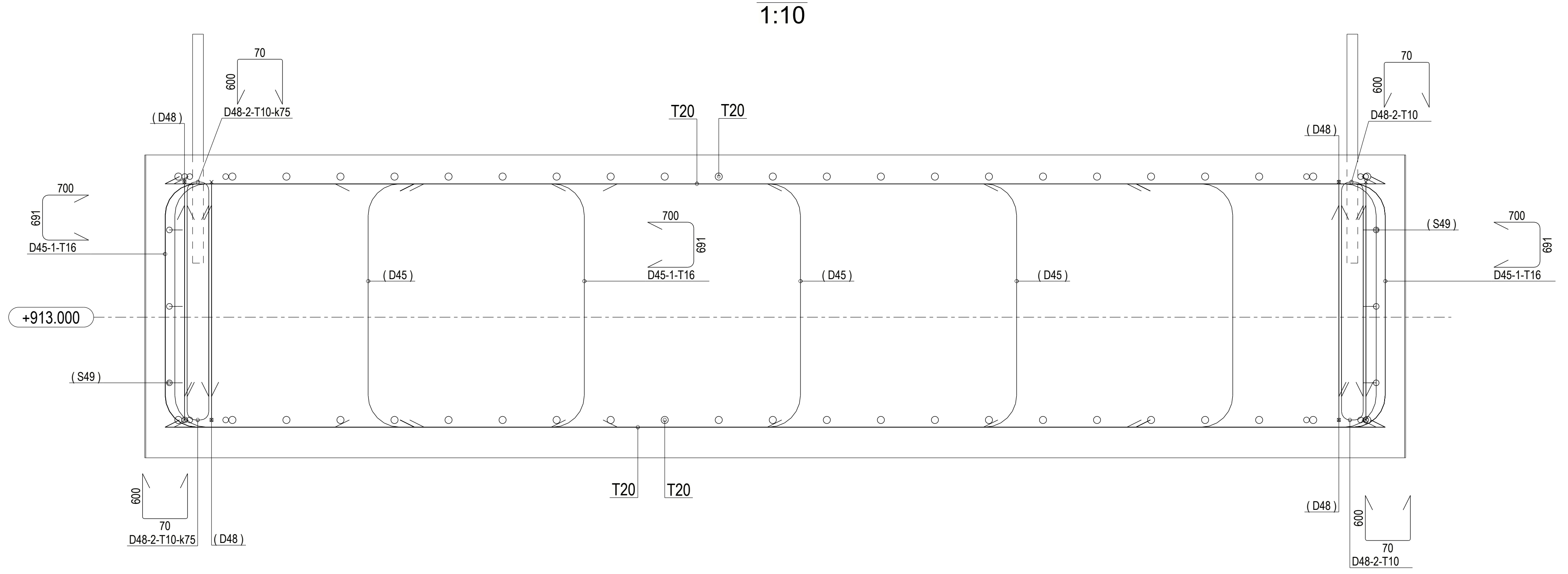
1:20


EQF/4
TL. + 913.607
2*D7-2-T10-k75
A1-6-T12- k300- L400
A57-3-T12- L2040
D3-15-T16- k150- L2350
A4-10-T16- k150- L2100
(D6)
(D5)
D56-3-T12
D59-3-T12
A57-3-T12- L2040
A1-6-T12- k300- L400
5*300
D
D

D - D
1:10


1316
D3-15-T16-k150
(A4)
(D7)
D7-2-T10-k75
(A57)
+913.000
A1-6-T12- k300- L400
HILTI HIT-HY 200
DRILL HOLE Ø14, h0=110 mm
A1-6-T12- k300- L400
HILTI HIT-HY 200
DRILL HOLE Ø14, h0=110 mm
556
480
70
D3-15-T16-k150
D7-2-T10-k75

B - B
1:10


(D48)
D48-2-T10-k75
D48-2-T10
T20
T20
D45-1-T16
D45-1-T16
(D45)
(S49)
+913.000
D48-2-T10-k75
D48-2-T10
700
600

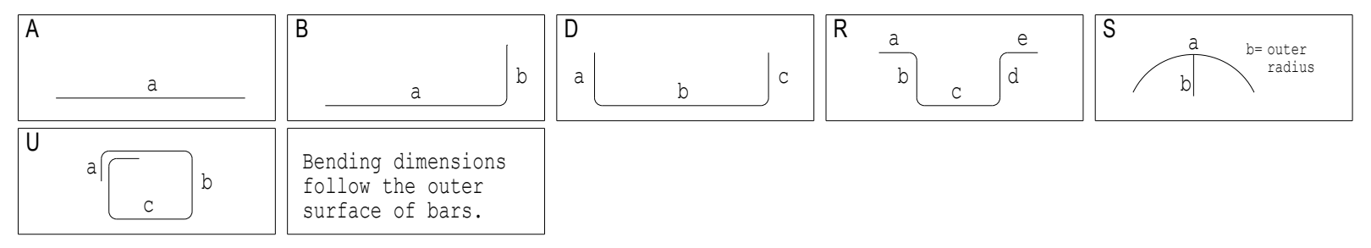
REINFORCING BAR LIST																	
CAST UNIT CODE: EQF/1																PCS:	1
REINFORCING BARS TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	BENDING DIMENSIONS [mm] a	b	c	d	e	u	v	x	TD	COMMENT
A	1	20	AISI304	12	400		7.1	398									
D	8	34	AISI304	16	1250		67.4	256	816	256						80	
A	9	12	AISI304	16	2350		44.6	2350									
D	10	6	AISI304	12	1480		7.9	642	254	642						60	
D	11	6	AISI304	12	1530		8.2	642	305	642						60	
D	12	32	AISI304	10	590		11.7	280	70	280						36	
D	16	32	AISI304	10	490		9.7	230	70	230						36	
D	22	4	AISI304	12	1910		6.8	606	756	606						72	
A	23	4	AISI304	12	2330		8.3	2330									
RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]:																	171.5

REINFORCING BAR LIST																	
CAST UNIT CODE: EQF/2																PCS:	1
REINFORCING BARS TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	BENDING DIMENSIONS [mm] a	b	c	d	e	u	v	x	TD	COMMENT
A	20	4	AISI304	20	1060		10.5	1064									
A	21	4	AISI304	20	1700		16.8	1700									
A	22	4	AISI304	20	2120		21.0	2122									
A	23	4	AISI304	20	2450		24.2	2449									
A	24	3	AISI304	20	2700		20.0	2701									
A	25	3	AISI304	20	2890		21.5	2895									
A	26	3	AISI304	20	3070		22.7	3068									
A	27	4	AISI304	20	3180		31.4	3179									
A	28	4	AISI304	20	3280		32.4	3284									
A	29	8	AISI304	20	3340		65.9	3337									
A	30	12	AISI304	20	3390		100.4	3387									
A	31	4	AISI304	20	3280		32.4	3278									
A	32	4	AISI304	20	3170		31.3	3169									
A	33	4	AISI304	20	3050		30.2	3052									
A	34	4	AISI304	20	2880		28.4	2879									
A	35	4	AISI304	20	2680		26.5	2678									
A	36	4	AISI304	20	2430		24.0	2427									
A	37	1	AISI304	20	1930		4.8	1930									
A	38	1	AISI304	20	1500		3.7	1504									
A	39	4	AISI304	20	990		9.8	991									
A	40	1	AISI304	20	2640		6.5	2640									
A	41	1	AISI304	20	2780		6.9	2783									
A	42	1	AISI304	20	2970		7.3	2968									
A	43	3	AISI304	20	2090		15.5	2091									
A	44	3	AISI304	20	1650		12.3	1654									
D	45	120	AISI304	16	1980		376.0	700	691	700						160	
D	46	2	AISI304	16	1870		5.9	700	580	700						160	
R	47	2	AISI304	10	860		1.1	190	130	320	130	190				60	
D	48	32	AISI304	10	1230		24.3	600	70	600						36	
S	49	6	AISI304	16	6080		57.7	6084	1690								
RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]:																	1071.2

REINFORCING BAR LIST																	
CAST UNIT CODE: EQF/3																PCS:	1
REINFORCING BARS TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	BENDING DIMENSIONS [mm] a	b	c	d	e	u	v	x	TD	COMMENT
A	1	17	AISI304	12	400		6.0	398									
A	2	17	AISI304	12	440		6.7	442									
A	13	38	AISI304	16	2450		147.1	2450									
D	14	19	AISI304	12	1720		29.0	600	576	600						60	
D	15	19	AISI304	12	1770		29.9	600	628	600						60	
A	17	35	AISI304	16	2650		146.5	2650									
D	18	18	AISI304	12	1770		27.9	600	626	600						60	
1	AT GROUP	18	PCS	...	1720	3	...	600	577	600							
D	19	18	AISI304	12	1770		27.9	600	626	600						60	
1	AT GROUP	18	PCS	...	1720	3	...	600	578	600							
D	50	32	AISI304	10	1130		22.3	550	70	550						36	
D	51	96	AISI304	10	1160		68.7	550	100	550						36	
U	52	30	AISI304	10	1010		18.8	140	215	215						60	
A	54	6	AISI304	12	2670		14.2	2670									
A	55	6	AISI304	12	2450		13.1	2450									
B	58	12	AISI304	12	970		10.3	500	500							72	
RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]:																	568.6

REINFORCING BAR LIST																	
CAST UNIT CODE: EQF/4																PCS:	2
REINFORCING BARS TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	BENDING DIMENSIONS [mm] a	b	c	d	e	u	v	x	TD	COMMENT
A	1	48	AISI304	12	400		17.0	398									
D	3	60	AISI304	16	2350		223.2	556	1316	556						80	
A	4	40	AISI304	16	2100		132.7	2100									
D	5	20	AISI304	12	1750		31.1	642	523	642						60	
D	6	20	AISI304	12	1800		31.9	642	567	642						60	
D	7	80	AISI304	10	990		48.9	480	70	480						36	
D	56	6	AISI304	12	2740		14.6	900	1280	600						32	
A	57	12	AISI304	12	2040		21.7	2036									
D	59	6	AISI304	12	2440		13.0	600	1280	600						32	
RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]:																	534.1

REINFORCEMENT BENDING SHAPES:


A
a
B
a
b
D
a
b
c
R
a
b
c
d
e
S
a
b
b= outer radius
U
a
b
c

Bending dimensions follow the outer surface of bars.

CONCRETE INFORMATION	
Concretes used	C35
Concrete cover 1	50±10mm

EMBED LIST
CAST UNIT WEIGHT IS CALCULATED USING CONCRETE VOLUME AND DENSITY 2500 kg/m3 + weight of embedded objects.

CAST UNIT POSITION	PCS		
EQF/1	1		
CONCRETE	NAME	VALUE	SUM
C35/45	EQUIPMENT_FOUNDATION	0.84	m³
	CAST UNIT TOTAL WEIGHT:	2.10	t
VALUE	UNIT	EMBEDS	
8	pcs	HIT-HY200-A+AM-HDG8.8 S355J2	
21.3	kg	AISI304 ø10	
38.2	kg	AISI304 ø12	
112.0	kg	AISI304 ø16	

EMBED LIST
CAST UNIT WEIGHT IS CALCULATED USING CONCRETE VOLUME AND DENSITY 2500 kg/m3 + weight of embedded objects.

CAST UNIT POSITION	PCS		
EQF/2	1		
CONCRETE	NAME	VALUE	SUM
C35/45	EQUIPMENT_FOUNDATION	8.04	m³
	CAST UNIT TOTAL WEIGHT:	20.11	t
VALUE	UNIT	EMBEDS	
4	pcs	HIT-HY200-A+AM-HDG8.8 S355J2	
25.4	kg	AISI304 ø10	
439.6	kg	AISI304 ø16	
606.3	kg	AISI304 ø20	

EMBED LIST
CAST UNIT WEIGHT IS CALCULATED USING CONCRETE VOLUME AND DENSITY 2500 kg/m3 + weight of embedded objects.

CAST UNIT POSITION	PCS		
EQF/3	1		
CONCRETE	NAME	VALUE	SUM
C35/45	EQUIPMENT_FOUNDATION	5.04	m³
	CAST UNIT TOTAL WEIGHT:	12.61	t
VALUE	UNIT	EMBEDS	
4	pcs	HIT-HY200-A+AM-HDG8.8 S355J2	
109.8	kg	AISI304 ø10	
165.1	kg	AISI304 ø12	
293.7	kg	AISI304 ø16	

EMBED LIST
CAST UNIT WEIGHT IS CALCULATED USING CONCRETE VOLUME AND DENSITY 2500 kg/m3 + weight of embedded objects.

CAST UNIT POSITION	PCS		
EQF/4	2		
CONCRETE	NAME	VALUE	SUM
C35/45	EQUIPMENT_FOUNDATION	3.97	m³
	CAST UNIT TOTAL WEIGHT:	4.97	t
VALUE	UNIT	EMBEDS	
10	pcs	HIT-HY200-A+AM-HDG8.8 S355J2	
48.9	kg	AISI304 ø10	
129.3	kg	AISI304 ø12	
356.0	kg	AISI304 ø16	

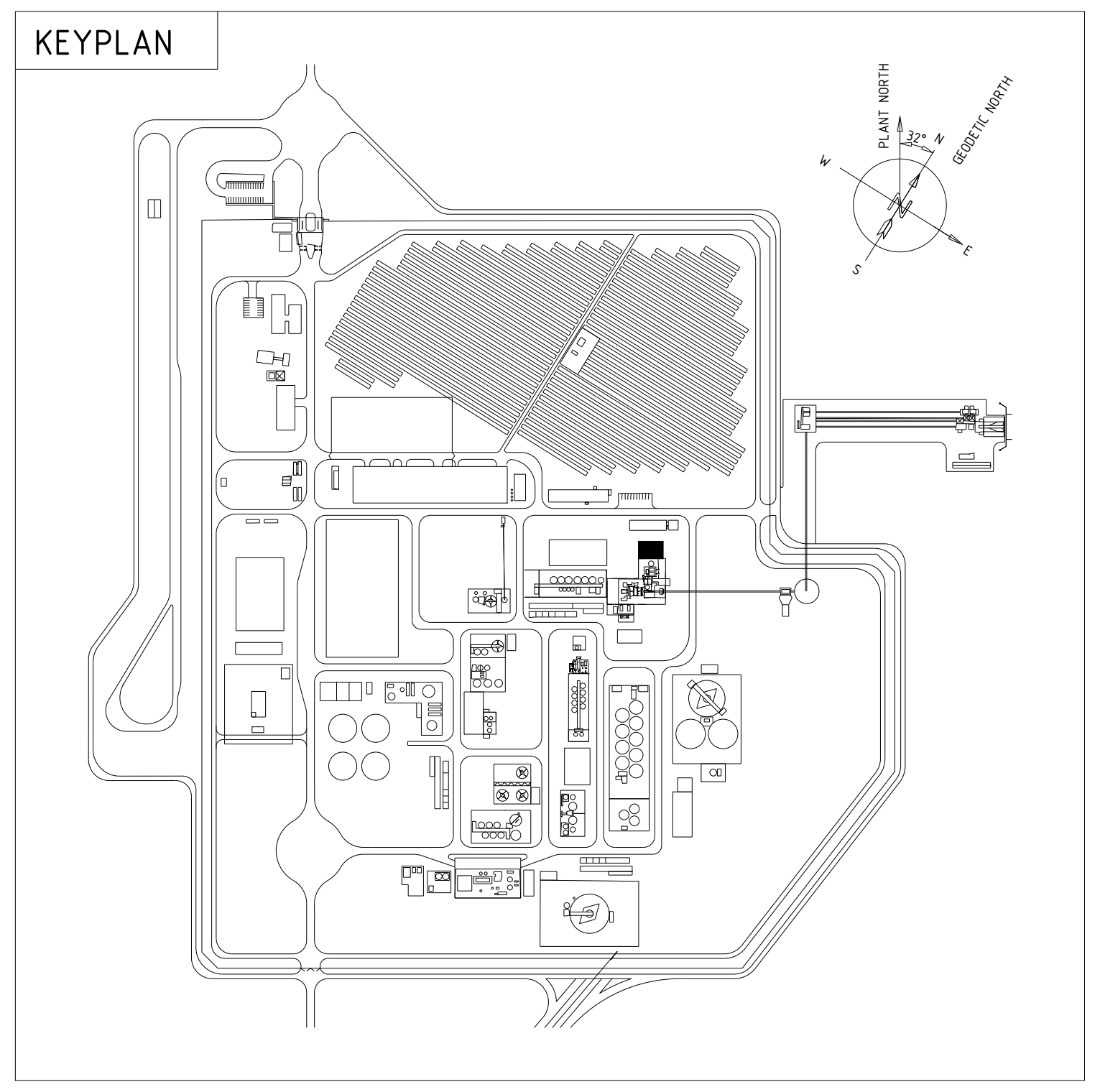
LEGEND:

R.L. - REDUCED LEVEL
T.O.C. - TOP OF CONCRETE
TL. - TOP LEVEL
B.O.F. - BOTTOM OF FOUNDATION
F.G.L. - FINISHED GROUND LEVEL
P.C.C. - PLAIN CEMENT CONCRETE
TYP - TYPICAL
MKD - MARKED
CL - CENTER LINE
BOTTOM REINF. (B)
TOP REINF. (T)
SBC - SAFE BEARING CAPACITY
MIN. - MINIMUM
REF. - REFERED
GA - GENERAL ARRANGEMENT
DWG. - DRAWING
R.C.C. - REINFORCED CEMENT CONCRETE

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS ARE IN METRES UNLESS OTHERWISE STATED
2. GRADE OF CONCRETE SHALL BE C35 (35MPa) UNLESS NOTED OTHERWISE & C15 (15MPa) FOR BLINDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615-GRADE 60 (fy=420MPa)
4. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS:
a) FOUNDATION: 75 mm (BOTTOM & SIDE)
b) FOUNDATION: 50 mm (TOP)
c) COLUMN & PEDESTAL: 50 mm
5. MINIMUM LAP LENGTHS OF REINFORCEMENT SHALL BE 40XBAR DIAMETER
6. FOR STANDARD REINFORCEMENT DETAILS & GENERAL NOTES FOR CIVIL WORKS REFER TO DRAWING NUMBER MD-216-1000-EG-CV-DED-1005 & 1001
7. ANCHOR BOLTS AND EMBEDDED STEELWORK SHALL BE HOT DIP GALVANIZED
8. T.O.C. & TOP OF SCREED TOLERANCE SHALL BE WITHIN +/- 10MM OF DESIGN LEVEL
9. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION IS CONSIDERED AS 375 KN/SQ.M.
10. REBARS CAN BE MOVED LOCALLY BY +/- 25mm TO AVOID INTERFERENCE WITH EMBEDDED PARTS

KEYPLAN


PLANT NORTH
GEODETIC NORTH
32°
N
W

DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3220-EG-CV-GAD-1154	DIMENSIONAL DRAWING		
MD-216-3220-EG-CV-CAL-1162	CALCULATION REPORT		

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	19.12.2023	AS-BUILT	NadSal	CemEce	NadSal	LuiPiz	
C01	22.02.2023	ISSUED FOR CONSTRUCTION	NadSal	CemEce	NadSal	LuiPiz	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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Saudi Arabian Mining Company شركة التعدين العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER
REVISION

OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707_GAU01_ZG01_CTB17_32211	OU100724167	1

DRAWN	NadSal	22.02.2023
CHECKED	CemEce	14.02.2023
ENGINEER	NadSal	22.02.2023
APPROVED	LuiPiz	14.02.2023
EMPLOYER		
SCALE	1:20 1:10	A0

MANSOURAH - MASSARAH GOLD PROJECT
REINFORCEMENT DRAWING
FOUNDATIONS
GRINDING MILL COOLING WATER

DRAWING NUMBER	REV
MD-216-3220-EG-CV-DED-1161	D01

Tekla Structures